Brandon Lee

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Education

New Jersey Institute of Technology (Newark, NJ) September 2021 - December 2025

Bachelor of Science - Computer Engineering

- 3.7/4.0 GPA
- **Relevant Coursework:** Data Structures & Algorithms, Programming Language Concepts, Operating Systems, Computer Architecture, Microprocessors, Signal Transmission, Digital Design, Circuits and Systems, Discrete Analysis, Differential Equations, Linear Algebra, Digital & Analog Circuit Lab, Computer Communications Lab

Professional Experience

Center for Solar-Terrestrial Research, NJIT (Newark, NJ) September 2023 - Present

Incoherent Scatter Radar Researcher

- Designed and programmed a **full-stack Python/Flask** simulation to model ISR signal transmission and backscatter with **interactive parameter controls** and **real-time visualization**
- Built modular numerical **computation pipelines** to process and analyze radar spectra under varying conditions
- **Refactored** and **integrated** multiple research scripts into a unified, reproducible framework, improving maintainability and experiment consistency
- Developed **data visualization** tools for spectral analysis, **parameter sweeps**, and **result comparison**; **automating workflows** to enable rapid experimentation and reduce manual configuration errors

Projects

- ***NJIT Electrical and Computer Engineering Senior Design Showcase First Place Winner:*** **Machine Learning Enhanced Myoelectric Prosthetic Arm**

Designed an **end-to-end signal processing** and **classification system** for real-time EMG-based gesture recognition. Implemented **feature extraction** and **Convolutional Neural Network-based machine learning classifiers** in **Python/MATLAB** for clear, low latency control; translating classified gestures into responsive motor action. Built **software modules** for data acquisition, preprocessing, classification, and control logic. Validated system performance through **real-time testing** and **iterative tuning**.

- **Incoherent Scatter Radar Website**

Designed a **centralized web platform** for consolidating ISR data from across the field for educational and research purposes as well as **performing simulations** to test backscatter response. Implemented routing for website navigation, SQL for information storage, and Python to calculate spectra and display response on custom UI.

- **Wireless Communication Hub**

Created a **distributed control system** enabling **wireless communication** between a central interface and peripheral devices using a custom **communication protocol** between an **embedded controller** and **MATLAB-based UI**.

Skills

- **Programming Languages:** Python, C++, C, Java, HTML, CSS, JavaScript, MATLAB, Assembly,
- **Software & Frameworks:** Flask, Git, Linux, Bash, SQL, NumPy, SciPy, React, Matplotlib, MATLAB ML Toolbox, scikit-learn, Plotly, Pandas, TensorFlow, Pytorch, REST APIs
- **Embedded & Low-Level:** Arduino, Microcontrollers, UART/I2C/SPI, Sensors, ADC/DAC, GPIO, PWM, Interrupts, Peripheral Interfacing